MILITARY SPECIFICATION

SIGHTUNIT: XM64E2

"This specification is approved for use by all departments and agencies of the Department of Defense".

1. Scope

1.1 This specification covers one type of instrument known as Sight-unit: XM64E2 used with the Lightweight Company Mortar, 60MM XM224 (See 6.1. Throughout this specification this instrument is referenced as "sightuni." This sightunit is composed of Telescope, Elbow: 11733780 and Mount, Telescope: 11741796. This sightunit has radioactively illuminated scales, level vials and reticle.

2., Applicable Documents

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-T-48574	Telescope, Elbow: 11733780
MIL-M-48575	Mount, Telescope: 11741796

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Conmander, ARRADCOM, ATTN: DRDAR-TST-S, Dover, N.J. 07801 bY using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1240

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MIL-STD-1458	Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification in Use Storage and Transportation

DRAWINGS

U.S. Army, Frankford Arsenal

F11733780	Telescope, Elbow: 11733780-
F11741726	Sightunit: 11741726
F11741796	Mount, Telescope: 11741796
F11750709	Adapter, Final Inspection, Azimuth

PACKAGING DATA SHEET

11741726	packaging of Sightunit: XM64E2

(Copies of specifications, standards, drawings, and packaging data sheet required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 cFRl71-190-	Interstate Commerce Commission and Departmrnt of Transportation Rules and Regulations for the Transportation of Explosives and other dangerous articles.

(Application for copies shouldbe addressed to the superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory)

Part 20:	Standards for Protection against radiation
Part 30:	Rules of General applicability to Licensing of by-product material.

(Applications for copies Should be addressed to the superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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3. REQUIREMENTS

3.1 Radioactive material .- prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, possession, handling and shipping of radioactive material.

3.2 Regulations pertaining to radioactive devices.- Marking, identification, handling and shipping of radioactive devices shall be in accordance with MIL-STD-1458.

3.3 Fabrication.- The sightunit shall be manufactured in accordance with Drawing F11741726 and drawings pertaining thereto.

3.3.1 Telescope.- The telescope shall be fabricated in accordance with Drawing F11733780 and drawings pertaining thereto, and shall meet the requirements of MIL-T-48574.

3.3.2 Mount.- The telescope mount shall be fabricated in accordance with Drawing F11741796 and drawings pertaining thereto, and shall meet the requirements of MIL-M-48575.

3.4 General specification,- The contractor shall be responsible for adherence to, and compliance with, the requirements of MIL-F-13926 listed below:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.5 Telescope clamping mechanism.- When locked, there shall be no rotation of the telescope within the inside diameter of the clamp when a 35 inch-pound torque is applied. The clamping mechanism shall not rotate about its axis where a 12 inch-pound torque is applied, When unlocked, the telescope and clamp shall rotate freely. This requirement shall meet at an ambient temperature between 60°F and 90°F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three sightunits plus three each of all items covered by SQAP. The sample shall remanufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractors, prior to submitting the sample to the Government, shall inspect the sample ti insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Governs the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “submission of Product” provisions of MIL-STD-105.

4.3.2 Lamination and tests.

4.3.2.1 Mount.- Themount’ (see 3.3.2) shall have been inspected and accepted in accordance with the inspection provisions contained in MIL-M-48575 and the applicable Supplementary Quality Assurance Provisions (SQAP’S) listed in the Technical Data package (TDP).

4.3.2.2 Telescope.- The telescope (see 3.3.1) shall have been inspected and accepted in accordance with the inspection provisions contained in MIL-T-48574 and the applicable SQAP’S listed in the TDP.

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4.3.2.3 Sightunit.- Examination and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in Tables I and II herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - 100:.PERCENT (100%) INSPECTION

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Telescope clamping mechanism	3.5	4.6.1

NoTE: ThE test in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- The right is reserved to reject any unit of product found defective during inspection whether the lot as a whole is accepted or rejected. Rejected units may be repaired or corrected and resubmitted for inspection with the approval of, and in the manner specified by the procuring activity.

4.4 Special sampling.

4.4.1 General.- Three sightunits shall be selected at random from each 100 produced. The samples shall meet the requirements and tests in Table II.

TABLE II

NO	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301 l	Fabrication	3.3, 3.3.1 & 3.3.2	Applicable drawings and' specifications - Visual
302.	General specification	3.4	MIL-F-13926 - Visual

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment,

4,5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible-with the “Test Methods and Procedures” specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and "Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming, Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4,.5.2.3.1 Telescope clamping mechanism.- The Adapter, Final Inspection, Azimuth F11750709 together with a suitable stand shall be utilized in performing the telescope clamping mechanism test. The stand shall have the necessary stability for positioning the adapter so that it maintains the mounting dovetail surface and dovetail axis of the mount in a vertical plane within plus or minus one degree.

4.5.2.3.2 Torque requirements.- The torque wrench type tester for testing the telescope clamping mechanism shall have a range of zero to fifty inch-pounds. The accuracy of the torque wrench type tester shall be five percent throughout its entire zone of excursion. The torque wrench shall be equipped With suitable adapters to accommodate testing of the appropriate assembly.

4.5.2,3.3 Wall target.- The wall target shall contain a vertical and horizontal crossline. It shall be positioned 6 to 12 feet in front of the sightunit under test.

4.6 Test methods and procedures.

4.6.1 Telescope clamping mechanism.- Conduct this test with the mount positioned on the testing equipment specified in 4.5.2.3.1. Position the target so that the intersecting point of the vertical and horizontal target lines are coincident to the geometric axis of the clamp for the telescope. Install the telescope in the clamping mechanism of the mount and lock the clamp by applying a force of 5 to 7 inch-pounds to the locking lever.. By means of the elevation and azimuth knobs adjust the line of sight of the telescope so that its vertical and horizontal reticle lines are superimposed on the image at the vertical and horizontal target lines. The

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telescope clamping mechanism shall conform to the requirements of 3.5 when the applicable torques are applied to the telescope body and the clamp for the telescope. Any rotation of the telescope within the clamp, or rotation of the clamping mechanism about its own axis shall be determined by the permanent displacement (lack of coincidence) of the telescope reticle lines with respect to the target image lines.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11741726. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The sightunit, XM64E2 is used for laying the Lightweight Company Mortar, 60MM, XM224 in deflection and elevation for direct and indirect fire modes.

6.2 Ordering data.- Procurement Documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).
(e) Provisions for First Article Testing.

Custodian:

Army-MU

Preparing activity:

Army - MU

Project No. 1240 -A705

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-S-48573(MU) Sightunit: XM64E2

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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MIL-S-48573(AR)
AMENDMENT 1
5 November 1984

MILITARY SPECIFICATION

SIGHTUNIT: M64

This Amendment forms a part of Military Specification MIL-S-48573(MU) dated 15 April 1977, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments anti Agencies of the Department of Defense.

PAGE 1

1.1, last sentence: Change “scales and level vials” to “scales, indices, and level vials.”

PAGE 3

3.2: Change “(see 4 and 6)” to “, see Sections 4.2 and 6.2.”

PAGE 4

“ 3.6: Delete in its entirety.

PAGE 6

4.4.2: Change “(see 4.3.3)” to “(see 4.3.2)”.

PAGE 8

4.6.1: Delete second sentence and substitute new second sentence:

“Install the telescope in the clamping mechanism of the mount and lock it in place by applying a force of 15 ± 1 inch pounds to the locking lever and a force of 10 ± 1 inch pounds to the wing nut.”

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1240 -A834)

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